

EM MID-2

FALL 2024
SOLUTION AND GRADING RUBRIC.

PROBLEM 1

FACE	Outward unit $\hat{n}$	Flux
$x=1$	$\hat{i}$	0
$x=0$	$-\hat{i}$	0
$y=1$	$\hat{j}$	16
$y=0$	$-\hat{j}$	-16
$z=1$	$\hat{k}$	0
$z=0$	$-\hat{k}$	0

0.5 points for each correct option.

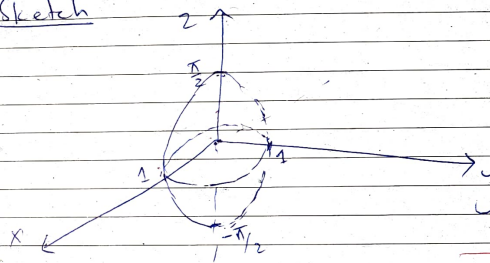
Instructor
comments
do not write
in this section

PROBLEM 2

$$x = \cos z, y = 0, -\frac{\pi}{2} \leq z \leq \frac{\pi}{2}$$

rotated around z-axis.

Sketch



0.5 for axis labels

0.5 for z & x intersect

0.5 for $x = \cos z$

0.5 for rotation

2 points for sketch

Parametrization

$$\vec{r}(\theta, z) = \langle \cos z \cos \theta, \cos z \sin \theta, z \rangle \rightarrow 1.5 \text{ points}$$

$$-\frac{\pi}{2} \leq z \leq \frac{\pi}{2}$$

$$0 \leq \theta \leq 2\pi$$

0.5 points

2 points for correct parametrization

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in this section

Finding normal

$$\vec{r}_\theta = \langle -\cos z \sin \theta, \cos z \cos \theta, 0 \rangle \quad (1)$$

$$\vec{r}_z = \langle -\sin z \cos \theta, -\sin z \sin \theta, 1 \rangle \quad (1)$$

$$\vec{N} = \vec{r}_\theta \times \vec{r}_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\cos z \sin \theta & \cos z \cos \theta & 0 \\ -\sin z \cos \theta & -\sin z \sin \theta & 1 \end{vmatrix}$$

$$= \langle \cos z \cos \theta, \cos z \sin \theta, \cos z \sin z \rangle \quad (1)$$

PROBLEM 3

Surface is the closed hemisphere given by
as the boundary of solid

$$x^2 + y^2 + z^2 \leq r^2, \quad z \geq 0$$

The boundary is piecewise smooth.
the curved hemisphere part given by

$$S_1: x^2 + y^2 + z^2 = r^2, \quad z > 0$$

and the base disk given by

$$S_2: x^2 + y^2 \leq r^2, \quad z = 0$$

1 Point for realizing the surface is
piecewise smooth and we need
to calculate two integrals.

S₁

Parametrization

$$\vec{r}(\phi, \theta) = \langle r \sin \phi \cos \theta, r \sin \phi \sin \theta, r \cos \phi \rangle$$

$$0 \leq \theta \leq 2\pi$$

$$0 \leq \phi \leq \pi/2$$

-1 point

$$\vec{r}_\theta = \langle -r \sin \phi \sin \theta, r \sin \phi \cos \theta, 0 \rangle$$

$$\vec{r}_\phi = \langle r \cos \phi \cos \theta, r \cos \phi \sin \theta, -r \sin \phi \rangle$$

$$\vec{r}_\theta \times \vec{r}_\phi = \langle -r^2 \sin^2 \phi \cos \theta, -r^2 \sin^2 \phi \sin \theta, -r^2 \sin \phi \cos \phi \rangle -1$$

$$|\vec{r}_\theta \times \vec{r}_\phi| = \left((-r^2 \sin^2 \phi \cos \theta)^2 + (-r^2 \sin^2 \phi \sin \theta)^2 + (-r^2 \sin \phi \cos \phi)^2 \right)^{1/2}$$

$$= \left(r^4 \sin^4 \phi \cos^2 \theta + r^4 \sin^4 \phi \sin^2 \theta + r^4 \sin^2 \phi \cos^2 \phi \right)^{1/2}$$

$$= r^2 \left(\sin^2 \phi (\sin^2 \theta + \cos^2 \theta) \right)^{1/2}$$

$$= r^2 \sin \phi -1$$

$$\text{Surface Area of } S_1 = \int_0^{2\pi} \int_0^{\pi/2} r^2 \sin \phi \, d\phi \, d\theta$$

$$= r^2 (2\pi) (1)$$

$$= 2\pi r^2 -1$$

S₂

Parametrization:

$$\vec{r}(u, \theta) = \langle u \cos \theta, u \sin \theta, 0 \rangle, \quad 0 \leq u \leq r, \quad 0 \leq \theta \leq 2\pi -0.5$$

$$\vec{r}_u = \langle \cos \theta, \sin \theta, 0 \rangle$$

$$\vec{r}_\theta = \langle -u \sin \theta, u \cos \theta, 0 \rangle$$

$$\vec{r}_u \times \vec{r}_\theta = \langle 0, 0, u \rangle$$

$$|\vec{r}_u \times \vec{r}_\theta| = u -0.5$$

$$\text{Surface Area of } S_2 = \int_0^{2\pi} \int_0^r u \, du \, d\theta$$

$$= \pi r^2 -1$$

$$\text{Surface Area of Closed Hemisphere} = \text{Surface Area of } S_1 + \text{Surface Area of } S_2$$

$$= 3\pi r^2$$

PROBLEM 4:

Sol.1 : $f(x, y, z) = (x^2 + y^2 + z^2)^{3/2}$
 $\vec{F} = \vec{\nabla} f$, find $\int_C \vec{F} \cdot d\vec{r}$

where $C: x^2 + y^2 = a^2$ oriented clockwise.

Since the field is gradient, it is conservative by the fundamental theorem of Vector Calculus

and thus $\int_C \vec{F} \cdot d\vec{r} = f(A) - f(A) = 0$

where A is any starting/ending point on C .

Sol.2: If some one has ^{correctly} calculated the line integral by finding $\vec{F}$ and parametrizing C , then they get full points.

Sol.3: Apply Stoke's Theorem

$$\vec{\nabla} \times \vec{F} = \vec{\nabla} \times (\vec{\nabla} f) = 0 \quad (\text{proven in HW2})$$

Thus $\oint_C \vec{F} \cdot d\vec{r} = \iint_S \vec{\nabla} \times \vec{F} \cdot d\vec{S} = 0$

where S is any smooth surface with C as its boundary.

PROBLEMS

Since no other information about the surface is known apart from this that it is piecewise smooth closed surface.

The only thing that can be done is to apply Divergence Theorem to find outward flux.

$$\begin{aligned} \oint_S \vec{F} \cdot \vec{n} \, dS &= \iiint_T \vec{\nabla} \cdot \vec{F} \, dV \quad \text{where } T \text{ is region enclosed by } S. \\ &= 8 \iiint_T dV \\ &= 8V \end{aligned}$$